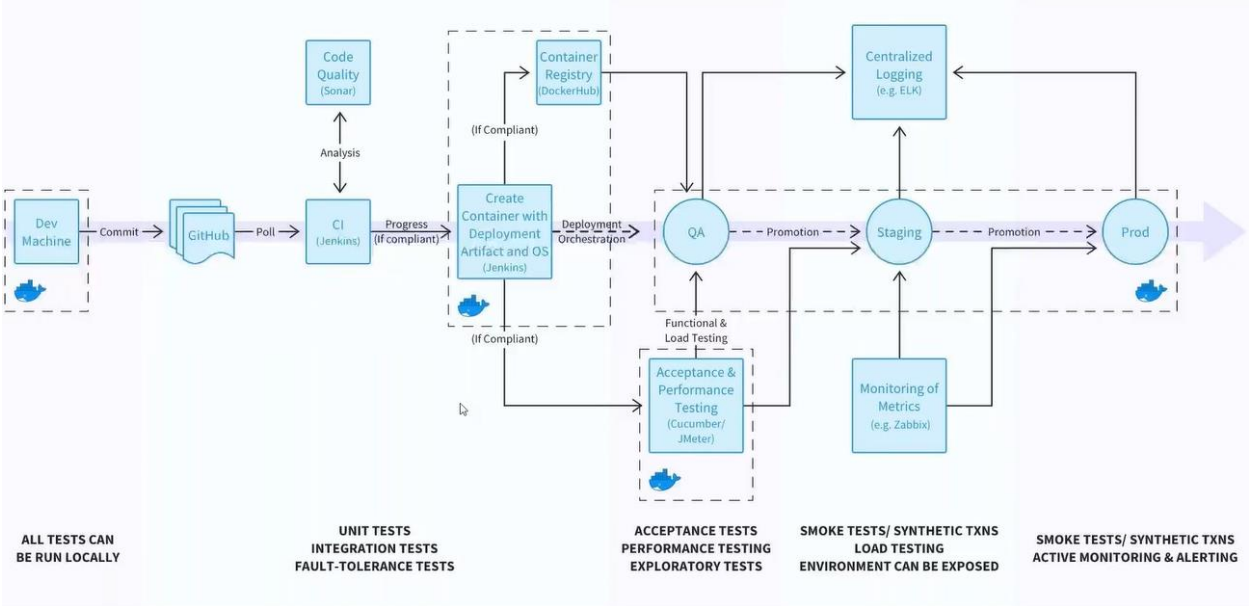
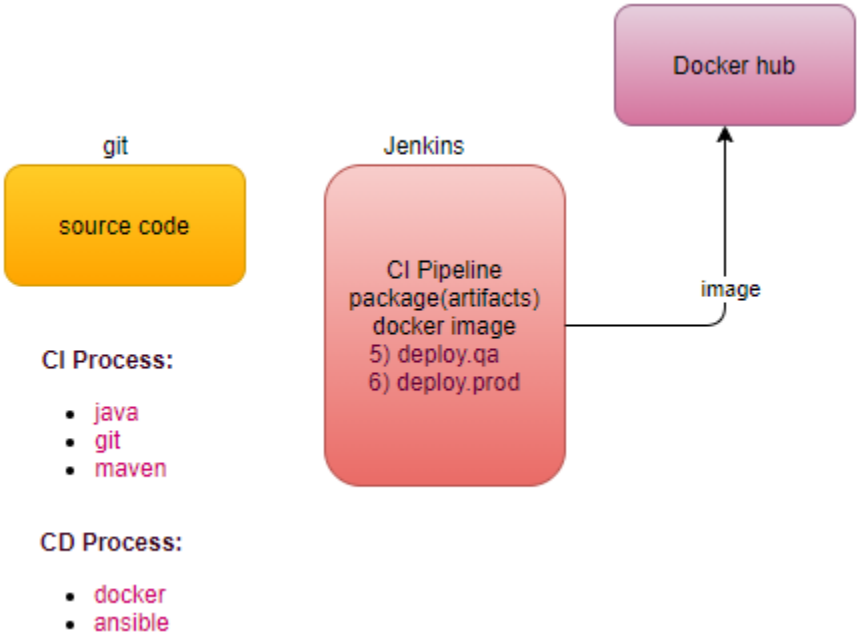


The output of CI => artifact (.war).

How to create docker image with the .war, and how to push it to the dockerhub

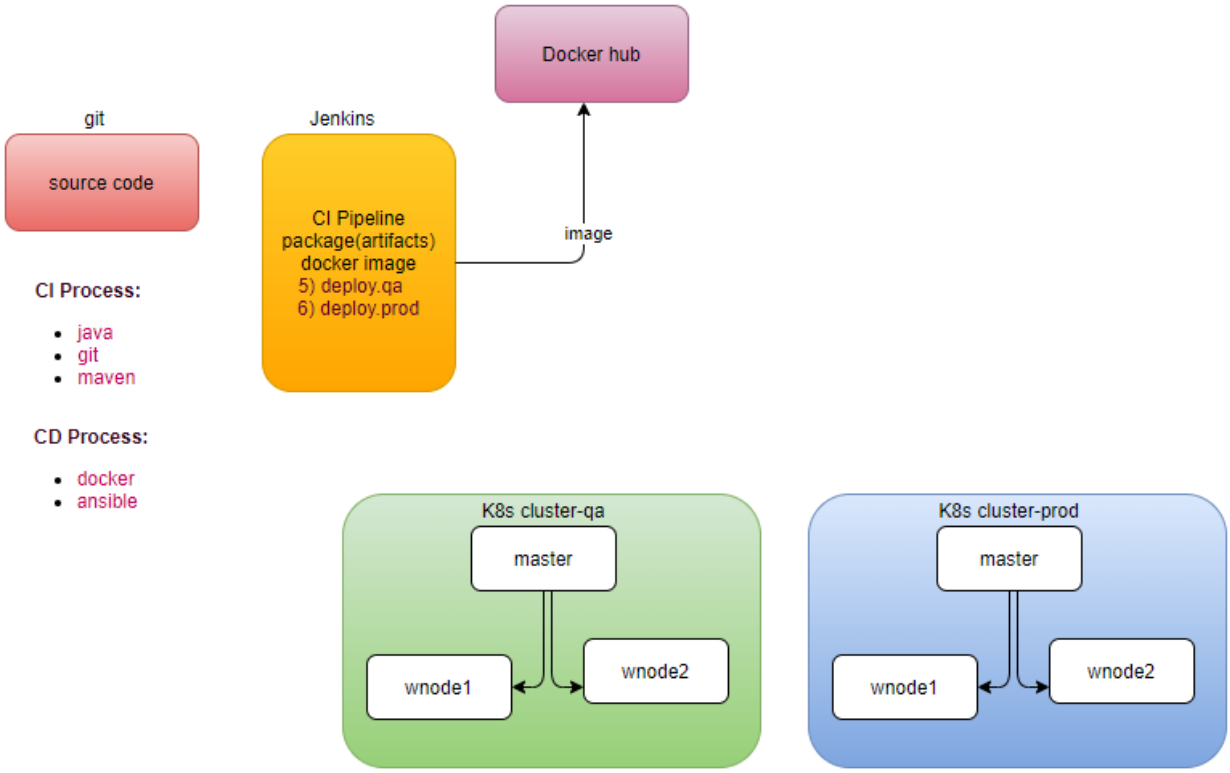


Continuous Delivery Process

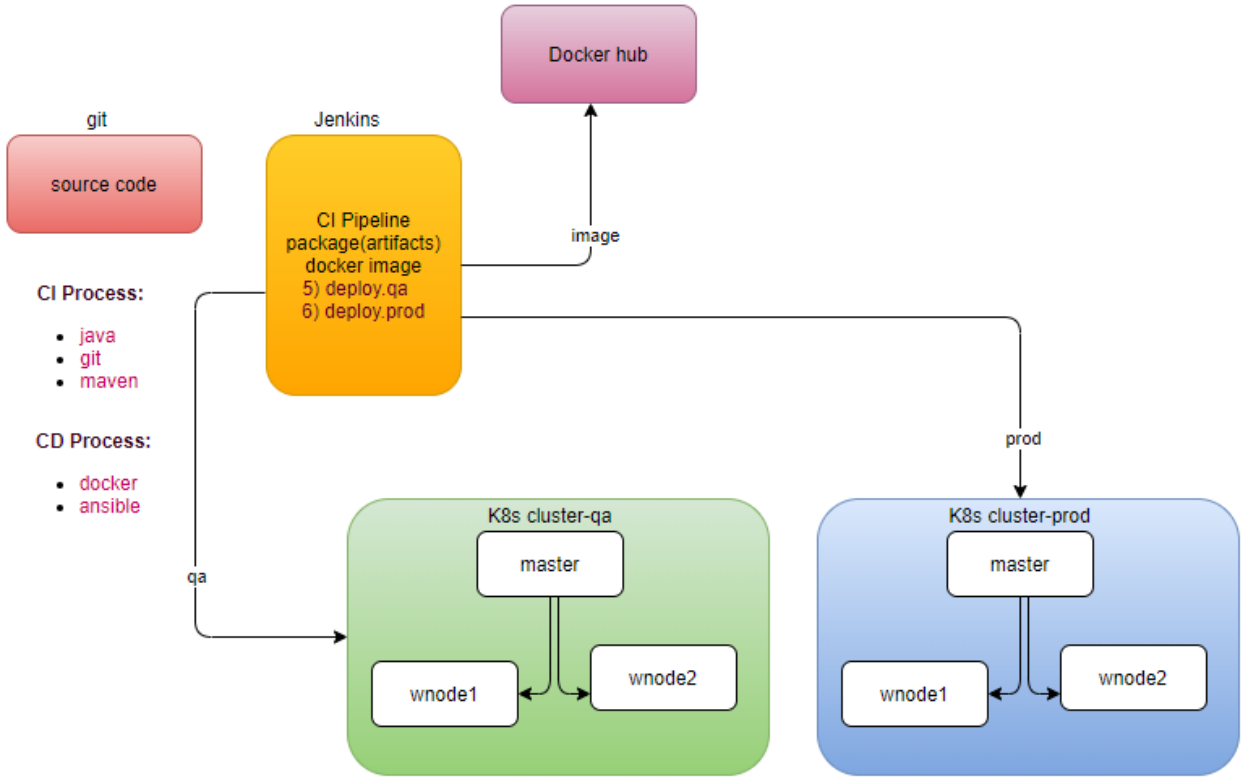
Integrate with docker -> create image

Integrate with ansible-> deploy in different environments (Environment is a group of servers).

pre-requisite-> Before deployment our environment must be ready.



Note: Jentink's its self doesn't login just trigger ansible to login into k8s master.



build docker image

step1: integrate docker with jenkins

step1.1: Install docker on the jenkins machine, just like we installed maven.

step1.2: Can jenkins run docker commands? Permission denied

step1.3: by default, jenkins runs the job as a jenkins user, whereas docker run as user root (access O/S kernel), means jenkins user doesn't have root privileges

```
root@jnode:/var/lib/jenkins/workspace# su - jenkins
jenkins@jnode:~$
jenkins@jnode:~$
jenkins@jnode:~$ docker ps
WARNING: Error loading config file: /var/lib/jenkins/.docker/config.json: open /var/lib/jenkins/.docker/config.json: permission denied
Got permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get http://%2Fvar%2Frun%2Fdocker.sock
/v1.24/containers/json: dial unix /var/run/docker.sock: connect: permission denied
jenkins@jnode:~$
```

step1.4: su - jenkins

docker ps # permission denied

sudo docker ps # no sudoers

sudo docker ps # ask for password

```
jenkins@jnode:~$ sudo docker ps
[sudo] password for jenkins:
jenkins@jnode:~$
```

step1.5: Give a Jenkins user sudoers privilege, **or** add jenkins user in the docker group.

/etc/sudoers

```
root@jnode:/var/lib/jenkins/workspace# cat /etc/sudoers
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "#include" directives:

#includedir /etc/sudoers.d
devops ALL=(ALL) NOPASSWD: ALL
root@jnode:/var/lib/jenkins/workspace#
```

visudoers

```
#includedir /etc/sudoers.d
devops ALL=(ALL) NOPASSWD: ALL
jenkins ALL=(ALL) NOPASSWD: ALL
~
~
```

service jenkins restart # service command always execute as a root user

now jenkins user can execute docker command as sudo

```
jenkins@jnode:~$ docker ps
WARNING: Error loading config file: /var/lib/jenkins/.docker/config.json: open /var/lib/jenkins/.docker/config.json: permission denied
Got permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get http://%2Fvar%2Frun%2Fdocker.sock
/v1.24/containers/json: dial unix /var/run/docker.sock: connect: permission denied
jenkins@jnode:~$
jenkins@jnode:~$
jenkins@jnode:~$
jenkins@jnode:~$
jenkins@jnode:~$
jenkins@jnode:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
jenkins@jnode:~$
```

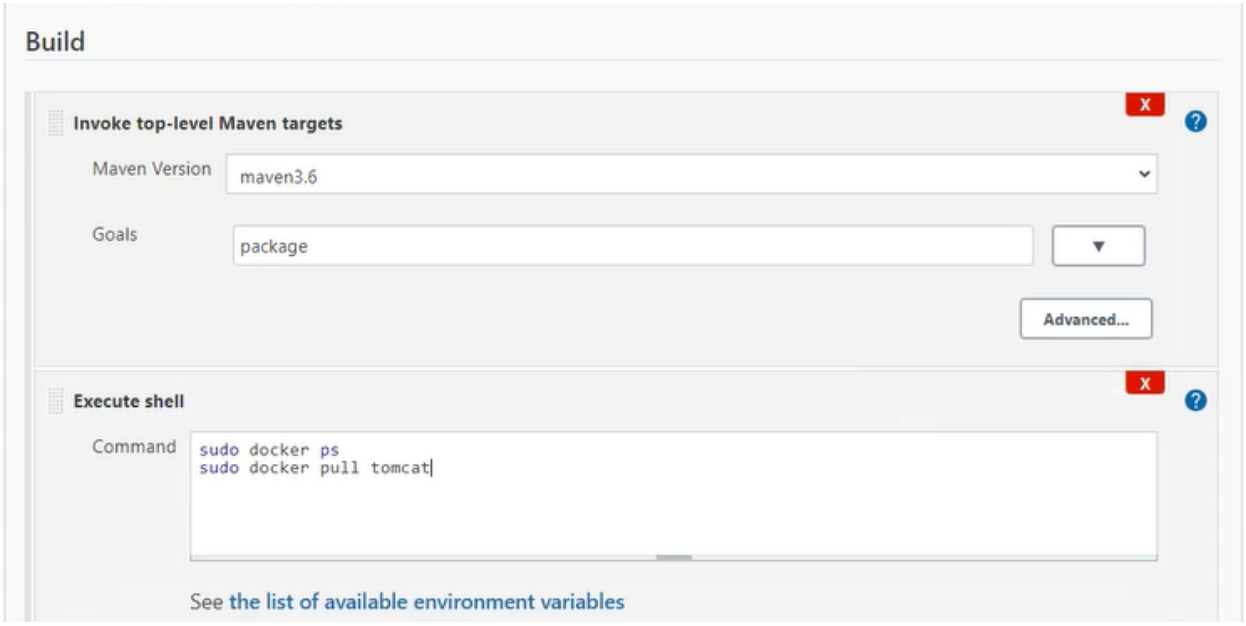
Validate:

re-login in to jenkins console

How to run docker commands from Jenkins's console?

(shell script)

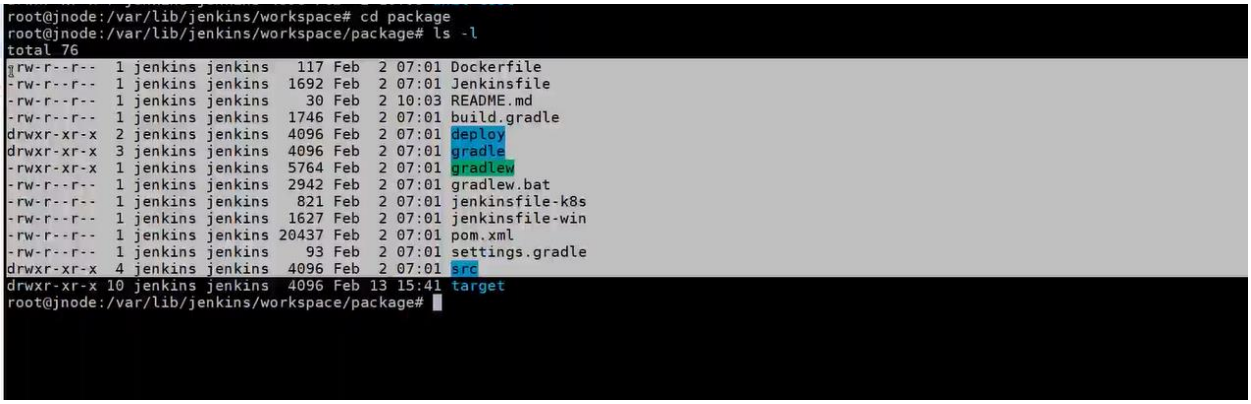
In the package job



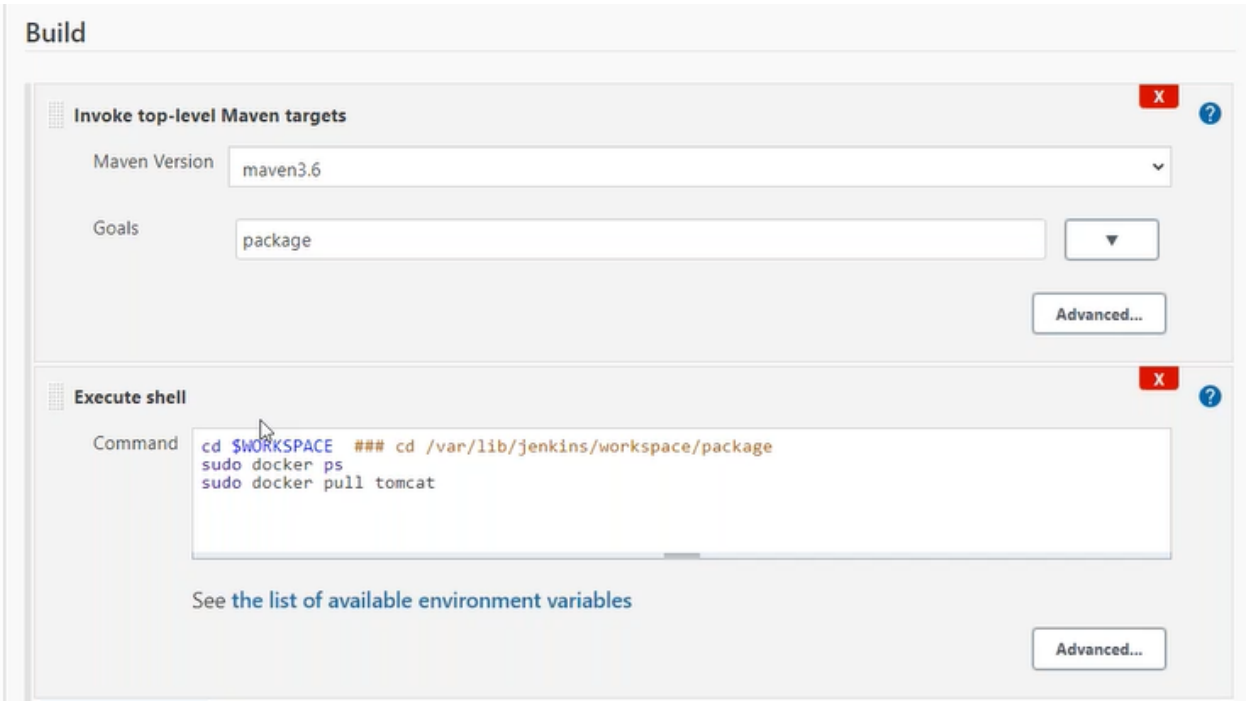
How to build an automated process?

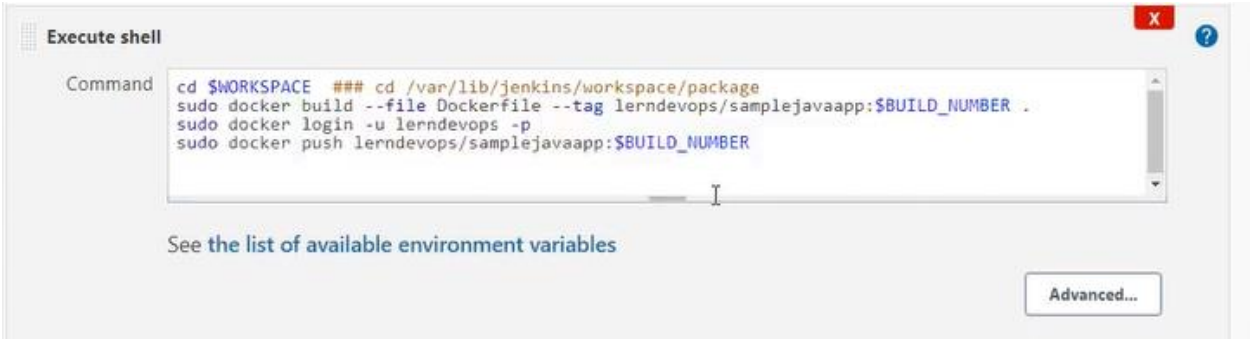
Dockerfile

Where docker clones the repository, under the workspace



step1: In the package job





/*

#!/bin/sh

WORKSPACE=/var/lib/jenkins/workspace/package

cd "$WORKSPACE"

$WORKSPACE ##cd /var/lib/jenkins/workspace/package

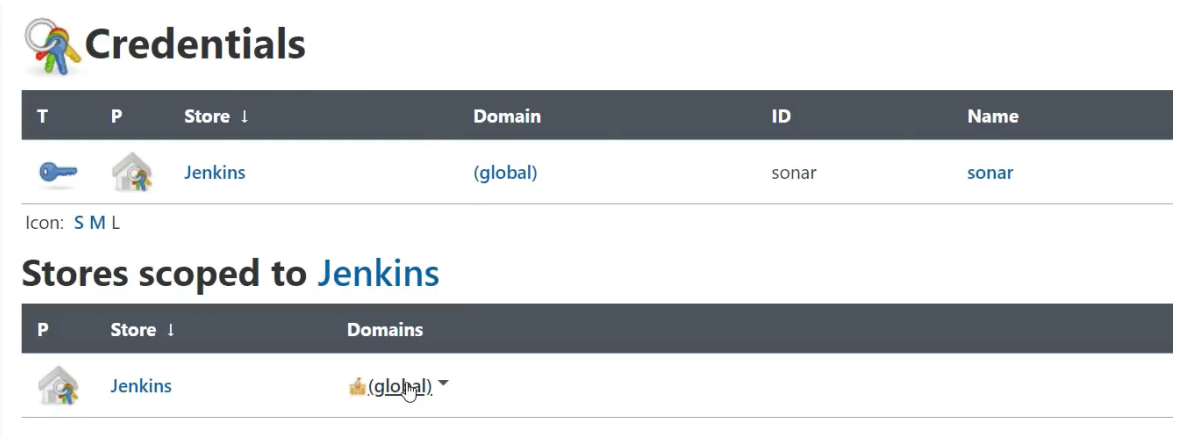
sudo docker build --file Dockerfile --tag neerajvohra/samplejavaapp:$BUILD_NUMBER .

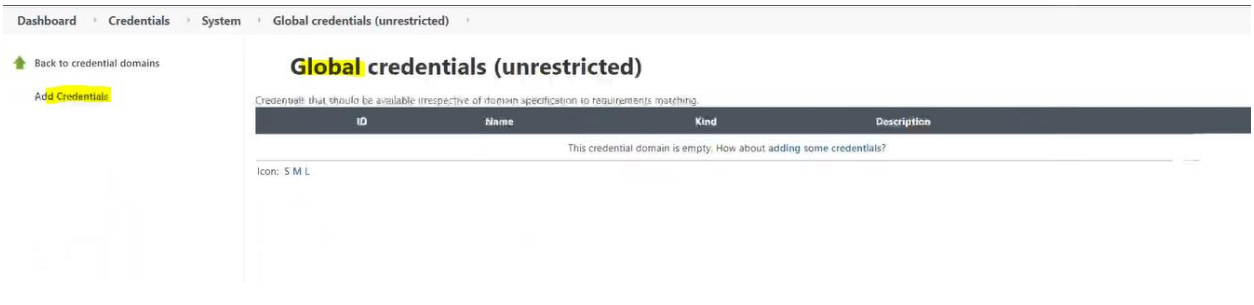
sudo docker login -u neerajvohra -p $Docker_Hub_Pass

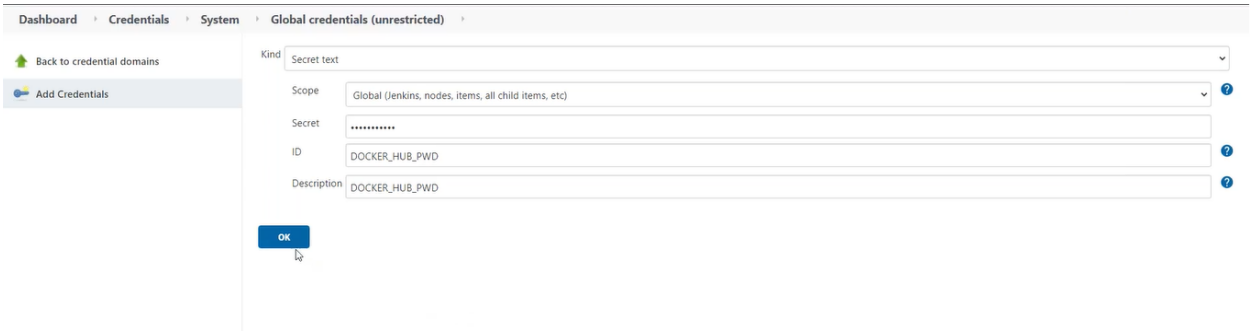
sudo docker push neerajvohra/samplejavaapp:$BUILD_NUMBER */

step1.1: How to encrypt password?

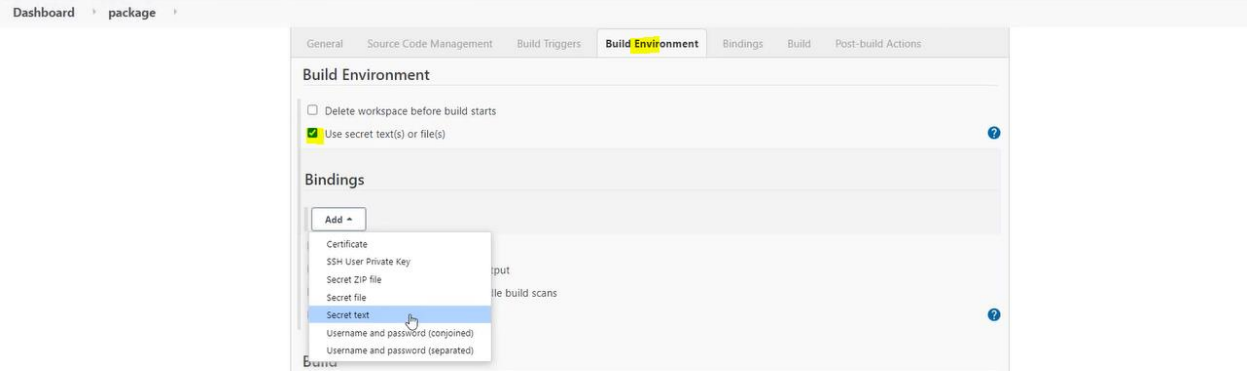
Manage Jenkins-> Manage Credentials # plugin Credentials







step1.2: How to get my credentials in my Job



How Integrate Ansible with Jenkins?

step1: setup the environment (k8s cluster)

step2: Jenkin can't log into master servers, will use ansible as a configuration tool to invoke the k8s client, we have to integrate ansible with jenkins

step2.1: If jenkins invoke ansible playbook or modules, install ansible on the same machine in which jenkins server is running.

step2.2: now we can call this machine is as an ansible controller, how controller identify the target servers (k8s master)? **Inventory**

step2.3:

su – devops # switch to devops user

vi myinv

```
[qa]
10.154.0.11

[prod]
10.128.0.29
```

step2.4: Now we need connection also, with username and password or username and sshkey. If we go with username and password then we have to again go with jenkins console.

step2.5: configure ssh keys

step2.5.1: generate ssh for ansible controller

ssh-keygen

ssh-copy-id devops@<ip of target masterqa>

ssh-copy-id devops@<ip of target masterprod>

step2.5.2: validate

ssh devops@<ip of target masterqa>

ssh devops@<ip of target masterprod>

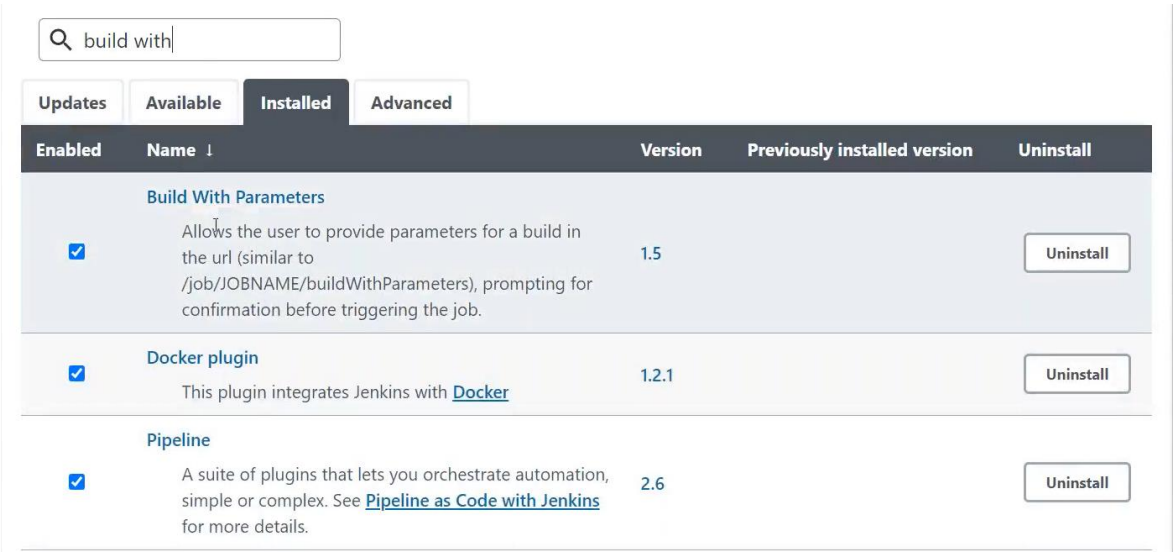
ansible all -i myinv -m ping

step3: create a new job -> freestyle deploy-qa

step3.1: where is my playbook?

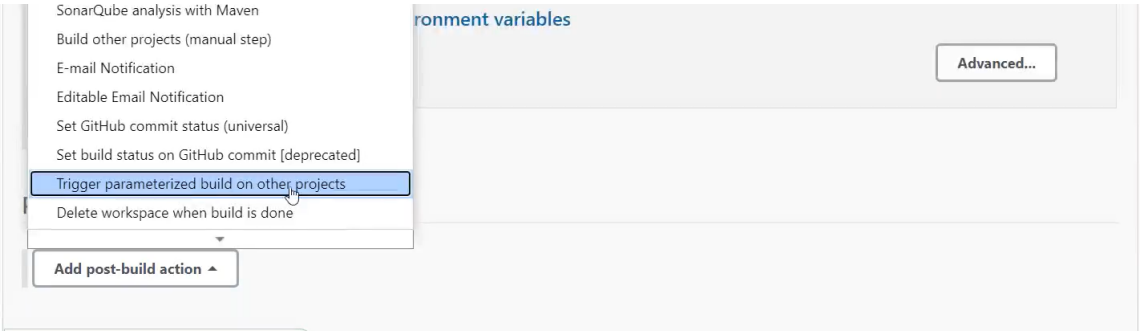
$BUILD_NUMBER: must be from previous section.

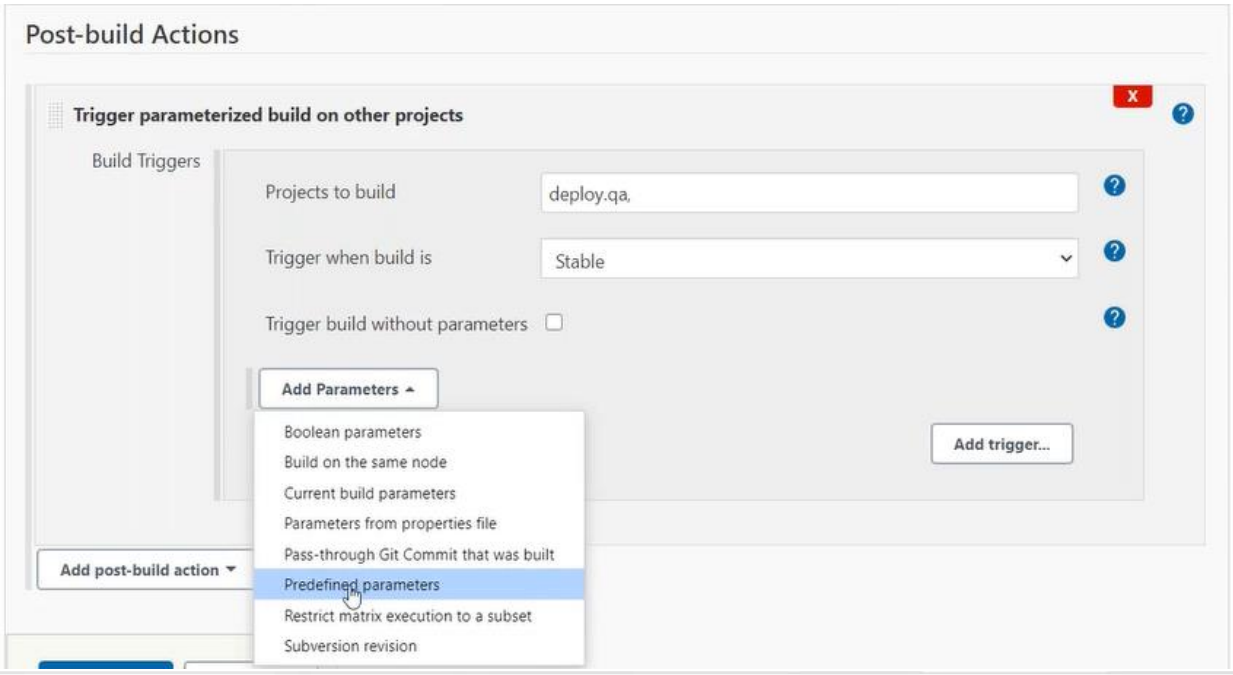
How to pass environment to next job? We have to add **plugin build with parameter** #CREDENTIAL_BINDING

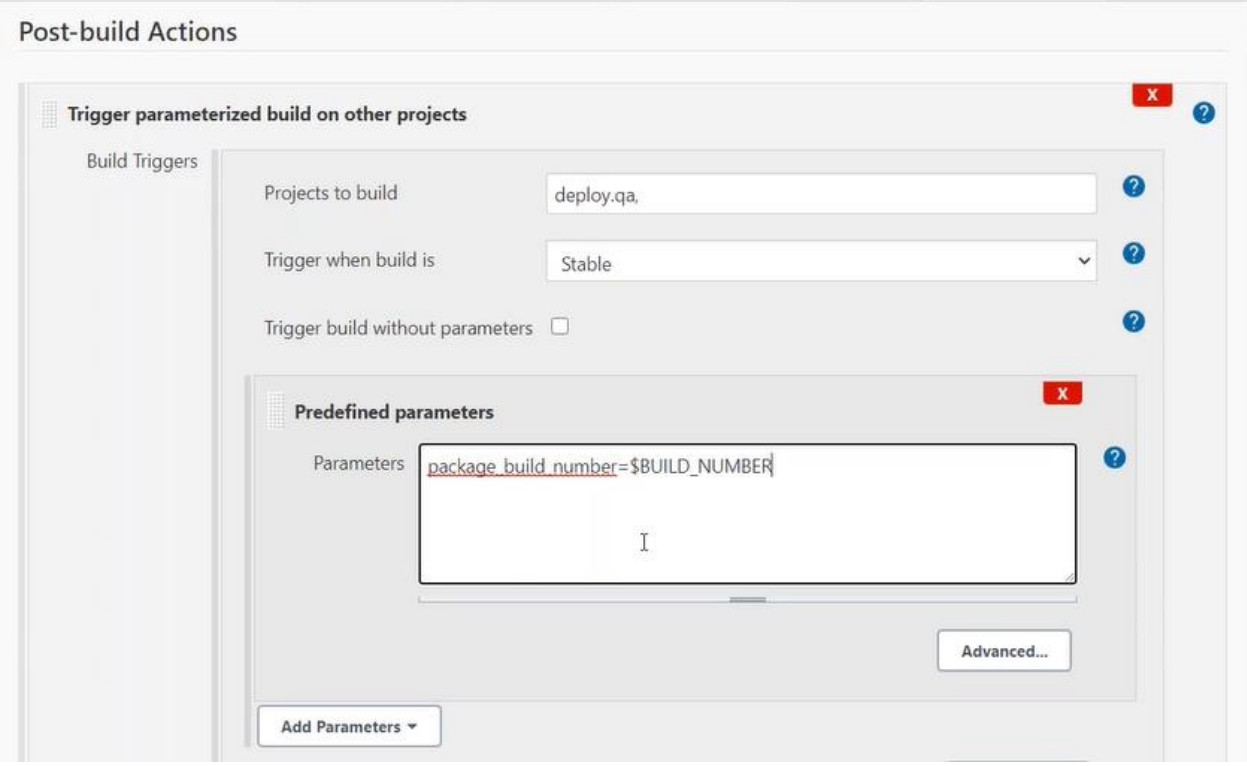


Step4: Go to the source job (package)

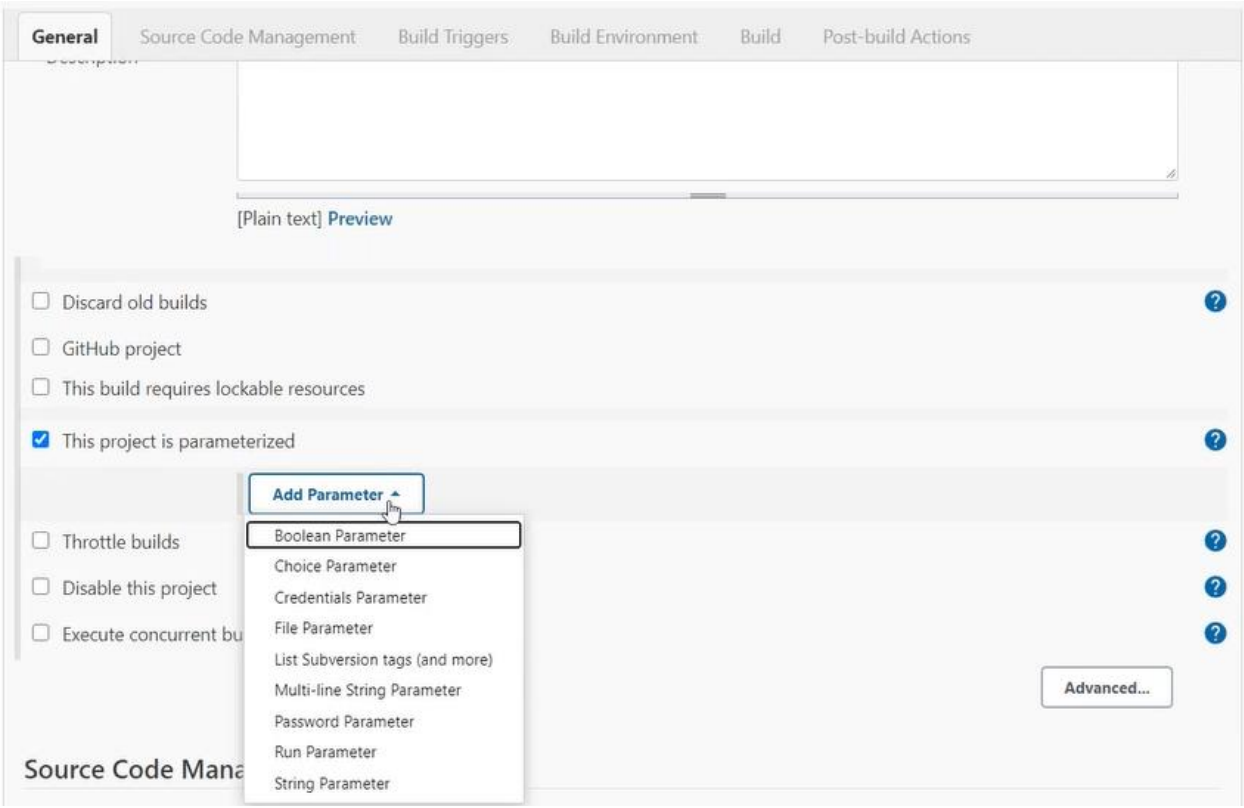
Add post-build-action-> Trigger parameterized build on other projects.

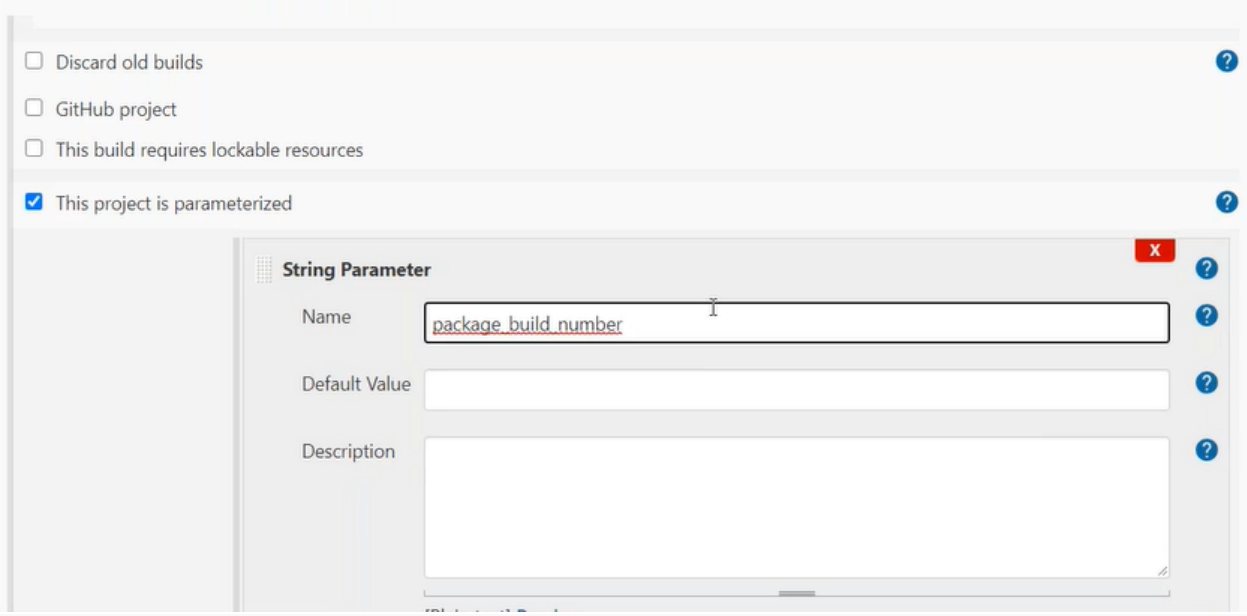



Post-build Actions
Trigger parameterized build on other projects
Build Triggers
Projects to build
deploy.qa,
Trigger when build is
Stable
Trigger build without parameters
Add Parameters
Boolean parameters
Build on the same node
Current build parameters
Parameters from properties file
Pass-through Git Commit that was built
Predefined parameters
Restrict matrix execution to a subset
Subversion revision
Add trigger...
Add post-build action


Post-build Actions
Trigger parameterized build on other projects
Build Triggers
Projects to build
deploy.qa,
Trigger when build is
Stable
Trigger build without parameters
Predefined parameters
Parameters
package_build_number=$BUILD_NUMBER
Advanced...
Add Parameters

Step5: Go to the target job (deploy-qa)





/*#!/bin/sh

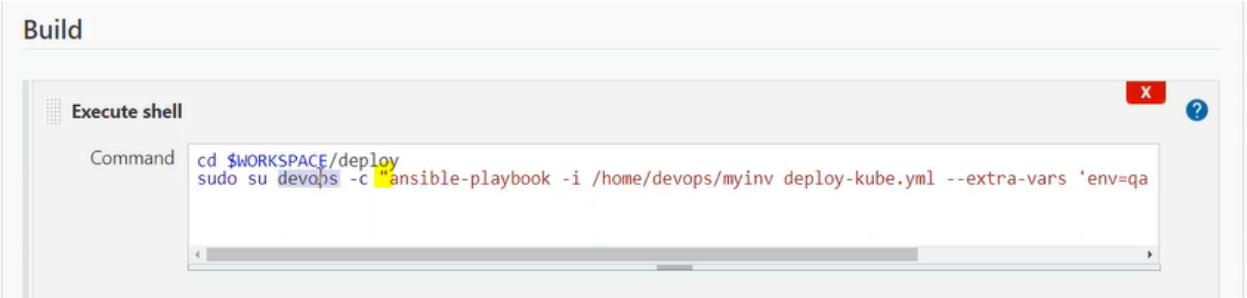
WORKSPACE=/var/lib/jenkins/workspace/ApplicationDeploy/deploy

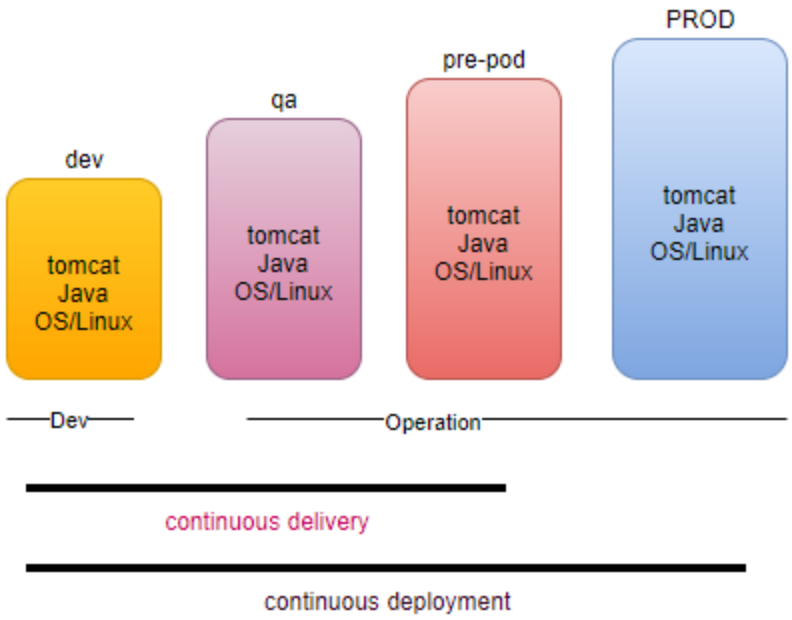
cd "$WORKSPACE"

sudo su devops -c "ansible-playbook -i /home/devops/myinv deploy-kube.yml --extra-vars='env=qa build=$package_build_number'" */

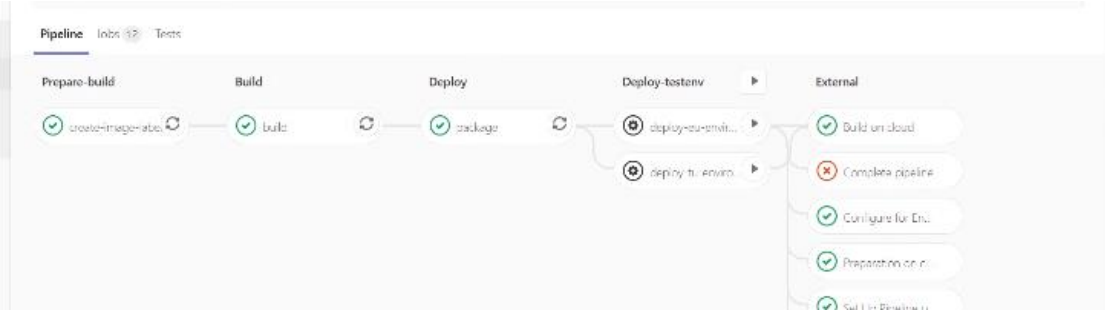
permission denied, ansible command is running as a jenkins user.

We have shared the ssh key for devops user but not for ansible user.





Continuous delivery is automated, with a single click our application available for QA.



How to create job deploy-prod, pass the variable from deploy-qa

Cloud provided devops tools.

DevOps OnPremise	DevOps on Cloud / AWS
Git/GitHub	code commit
Jenkins (maven)	codebuild/codepipeline/codedeploy
Ansible/Puppet	Cloud Formation
Docker	Docker
dockerswaarm/k8s	ECS / EKS
Nagios	cloud watch

- DevOps is a mindset, adapt the changes quickly and work on that.
- It’s a collaboration with various team. The way you implement, using various tools.
- Deliver your application in the market quicker ASAP.
- Adapting the new tools and technologies.